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CREATION AN ONTOLOGY OWL FOR E-BUSINESS SERVICES USING NATURAL LANGUAGE PROCESSING

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Área de Concentração: Ciência da Computação.

Orientador: Prof. Marco Aurélio Mendes

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I offer this work to all the girls that during childhood forever changed the dolls by the toolbox of dad.

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ABSTRACT

An ontology is responsible for formalizing shared concepts. For opinion mining, an ontology of product aspects shows the characteristics of the products making it accessible so that algorithms can identify users's opinion about aspects of the products evaluated on the web. For users and consumers of web products, having access to another user's review about a product you want to purchase is of great importance. That's the kind of information that influences on decision making, at the time of purchase. This research presented a methodology for the creation of an ontology of product aspects for e-business services. The ontology was implemented in OWL language. By using a set of opinionated texts containing user opinion about web products, we found that the formalization of knowledge about aspects of the evaluated products can enhance the e-business services. In addition, in the context of the semantic web, it is a necessary activity so that software agents can capture that information and make it available to users for decision-making at the time of purchase. The methodology uses natural language processing methods. We opted for this approach in terms of using user reviews of electronic products. These views are expressed on social networks in the form of text, or unstructured data. The PLN has a set of advanced techniques for the processing of such data. However, it is still necessary to investigate techniques linguistically enriched to make the process more efficient. The construction of the ontology aspects performed in this study, said several PLN methods applied concomitantly. I believe that for future work, a hybrid approach that also uses statistical models can bring better results. For the area proposed in this study achieved a 93% coverage. A good result, given that our proposal was the partial automation of the ontology construction process. Therefore, the implementation of ontologies products aspects to improve the e-business services in the making of web users decision at the time of purchase.

RESUMO

Uma ontologia é responsável por formalizar conceitos compartilhados. Para a mineração opinião, uma ontologia de aspectos de produtos é composta por características dos produtos, tornando-o acessível para que os algoritmos possam identificar o alvo da opinião os usuários dos produtos avaliados na web. Para usuários consumidores de produtos web ter acesso a avaliação de outros usuários sobre um produto é muito importante, pois esse é o tipo de informação que influencia na tomada de decisão, no momento da compra. Esta pesquisa apresenta uma metodologia para a criação de forma semiautomática de uma ontologia de aspectos de produtos que servirá de recurso para serviços de e-business. A ontologia foi implementada usando a linguagem OWL. Usando um conjunto de textos opinativos que continham opiniões dos usuários sobre os produtos web, descobrimos que a formalização do conhecimento sobre aspectos dos produtos avaliados pode melhorar os serviços de e-business. Além disso, no contexto da web semântica o produto desenvolvido neste trabalho gera informação estruturada para que agentes de software possam capturá-las, tornando-as disponível à usuários web. A metodologia adotada utiliza métodos de processamento de linguagem natural. Optamos por essa abordagem em função de utilizarmos opiniões de usuários sobre produtos eletrônicos. Essas opiniões são expressas nas redes sociais em forma de texto, ou seja dados não estruturados. A PLN possui um conjunto de técnicas avançadas para o processamento deste tipo de dados. Entretanto, ainda é necessário investigar técnicas enriquecidas linguisticamente para tornar o processo mais eficiente. A construção da ontologia de aspectos realizada neste trabalho, contou vários métodos de PLN aplicados concomitantemente. Acredito que para trabalhos futuros, uma abordagem híbrida que utiliza também modelos estatísticos poderá trazer melhores resultados. Para o domínio proposto neste trabalho alcançamos uma cobertura de 93%. Um bom resultado, tendo em vista que a nossa proposta foi a automatização parcial do processo de construção da ontologia. Além disso, concluímos também que a implementação de ontologias de aspectos melhoram os serviços de e-business no âmbito da tomada de decisão dos usuários web no momento da compra.

LIST OF ABBREVIATIONS

NLP - Natural language processing

OWL - Web ontology language

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1.0 INTRODUCTION

According to a survey in Lohse et al. (2000) the people buy in eletronic commerce to suffer from lack of time ("time starved," according to the authors). For those users do a manual analysis of criticism about e-commerce products recently released can take months, and therefore affect the decision-making process. In this context, the opinion mining is the area that automatically evaluates the opinions and sentiments expressed in textual form regarding a particular topic of interest (Carvalho, 2014). One of the approaches to opinion mining is aspect-based. In this approach, the opinions are divided into ’characteristics’ or ’aspects’ of the product evaluated. From the characteristics of the selected entities, the opinion of the user is taken, which can be divided into three aspects: positive, negative and neutral. The opinions are typically qualifiers of the entities, i.e. the users generally evaluates the characteristics of the products and not only the product itself. For example, in the sentence "*the voice quality of this phone is incredible*". We have the entity “*telephone”* and the evaluated aspect is the quality of the voice. And to this end we have a positive assessment of this aspect which is “*amazing”* (Liu, 2012). However, the identification of web products aspects is not an easy

task. In this work will be implemented a ontology OWL by to map and to share knowledge about e-commerce products aspects. Unlike in the area of artificial intelligence and knowledge engineering, that focus on creating ontologies abstract, e.g. WordNet and CyC, the Web of the future will be made up of several small ontologies and highly contextualized, developed locally by software engineers and ontologies engineers (Hendler, 2001). According to Fensel (2001) the Web community believes that, in the near future, all business on the network should provide the semantics of your pages through an ontology. A ontology is an explicit specification of a conceptualization (Gruber, 1993). The conceptualization refers to the meaning of concepts and its relationships, given the context of the domain. In the process of construction of ontology it is possible to get the domain of knowledge from various types of entries: texts, dictionaries, knowledge bases, semi structured schemes or database. In this work, we will use opinionated web texts. A default language, frequently used for the implementation of ontologies for the web is the OWL. The Ontology Web Language (OWL) is a declarative language.

1.1 Motivations

The main motivation in this work is to investigate methods and resource automated that enhance the making of decision at the time of purchase of web products.

1.2 Goals

We goals are to develop an ontology owl with web product aspect and to present a methodology proposal for semi-automatic domain ontology creation from web users reviews.

1.3 Expected results

Our expectation are to contribute to e-business service. The development of an ontology owl composed of user reviews on web products is an important because, as well as share

knowledge about a domain, can help consumers products web users in making of decision at the time of purchase.

2.0 FUNDAMENTALS

2.1 E-business services and opinion mining

The volume big of information a available in the web is necessary to develop computational resources responsible for automation the web service use. In this context, the e-business service are process or agents responsible for provide wide variety of services surrounding the goods, e.g. after sales services and information services, are developed in order to provide a better customer experience and a more personalized/customized environment (e.g. Pine and Gilmore, 1999). The opinion mining is a kind e-business responsible for evaluating, automatically, content expressing opinion of users on the web, about a target or reference. An opinion can be defined by identifying two elements of a document: the target of the opinion and the feeling expressed on this subject (Liu, 2012). A document is any piece of text in natural language Tsytsarau and all (2012). A target is represented by an entity or object or aspect. Aspects represent properties or parts of entities that are evaluated by users in opinionated texts (such as comments and blogs on the Internet) (Pang and Lee, 2008). Since

Web 2.0 has become a common practice, as users can freely express their feelings and sentiments about products, politics, sports and any other aspects that these users want.

2.2 Semantic web

The main semantic web goal is share data and to provide provide specialized semantic services for people. The Semantic Web was devised by Tim Berners-Lee, who also created concepts such as WWW, URIs, HTTP and HTML [3]. The Semantic Web is an extension of the current Web 2.0 whose main objective is to formalize the web content to make it understandable to both humans and machines. One of the main challenges of the current semantic web is to create a hybrid language capable of expressing the semantics of web content and concurrently set rules for processing of this meaning in order to infer new data and rules. These rules for the semantic processing are transferred to the web and allow software agents to interact with this content. The Semantic Web is represented by a three-tier architecture. The layer schemes 6 is responsible for structuring the data, logically and assign meaning to them, so that these data become accessible to web systems. They are used in this RDF and XML layer. The logic layer is responsible for thinking and execution of logical inferences from the previously described semantics and the application layer allows interaction between the user and the Semantic Web application. The ontology layer is responsible for interpreting the RDFS layer, providing a higher level of expressiveness to define the semantics of information. The recommends that people build ontologies using OWL. In this work we goal is create an ontology owl that will serve as resource in the semantic web

2.3 Ontology

Ontology is a concept originally used in philosophy to set objects of nature and relationships of beings. The word ontology comes from the Greek, which means study of being. In computing, we use the concept of ontology as defining a data model representing a vocabulary of a domain. From a fashion point of view, an ontology is a (Cimiano, 2006) structure (as shown figure 1): Where C is the set of concepts; R is the set of relationships; A are identifiers and attributes T data types.

$$\mathcal{O} := (C, \leqslant_C, R, \sigma_R, \leqslant_R, A, \sigma_A, T)$$

Fig.1 Ontology structure

Fonte: Cimiano, 2006

According to Jacob (2003) ontologies are categories of things that exist or may exist in a particular private domain, producing a catalog where there are relations between the types and even domain subtypes, providing a common understanding and shared knowledge of a domain that can be communicated between people and application programs. The role of ontologies is to explain the vocabulary used and provide a standard for information sharing.(Mazieiro and all, 2008). In an ontology, concepts are defined as classes, relationships between concepts are binary relationships named properties and also define the individuals who are members of classes. According to W3C, an ontology defines terms used in the description and in the representation of the area of knowledge. The process of generating Ontologies (Ontology Generation) or extracting Ontologies (Ontology Extraction) or also known as learning of ontologies (Ontology Learning) is defined as a set of methods and techniques for the construction of an ontology or 7 enrichment of an existing [9]. According to Maedche (2001) ontology learning approaches can be classified according to the entries that include: a) texts: starting from a corpus of a domain and using techniques of natural language processing and machine learning to learn terms and relationships; b) dictionaries: use the definitions of the lexical entries and use the existing relations between them (e.g. synonymy); c) knowledge bases: use the existing rules in the knowledge bases; d) semi-structured schemes: use predefined data structures and relationships as existing in eXtended Markup Language (XML); and e) relational databases: extract relevant concepts and relations of knowledge represented in the records and the schemas of the databases.

3.0 RELATED RESEARCH

Freitas and Vieira (2013) present a approach based features level opinion mining for portuguese. In this work the autores shows an opinion mining and sentiment analysis

architecture using ontology. A representative work in the area of creation of ontology from texts for representation of the opinion is to (Acir and al, 2007). This work presents the ontology representation for e-commerce products aspects, from opinionated texts on the web. The authors demonstrate that the ontology of e-commerce products aspects is also fundamental to recommendation systems of service. According to Acir and al (2007) the process for creating the ontology for product aspects is shown in figure 2:

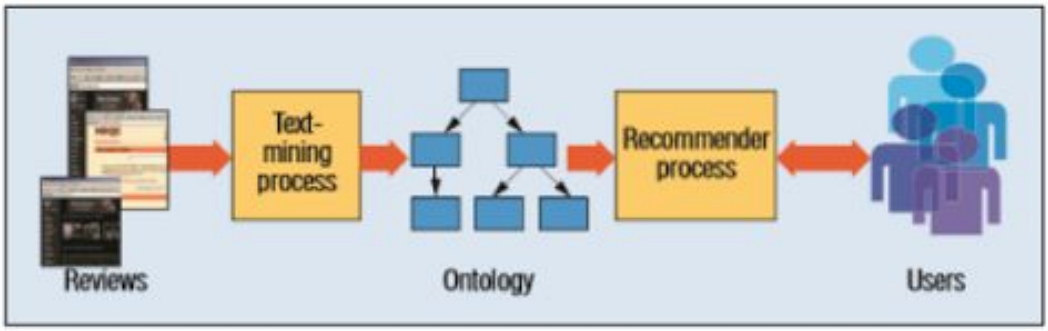

Fig.2 The structure of the process creating ontology
Fonte: Acir and al, 2007

Interestingly, the construction of an ontology e-commerce products aspects can be used in the e-business services, semantic web, opinion mining and recommendation systems. According to the work of Buitelaar (2005) the task of construction of ontologies, the texts comprises basically the following steps using natural language processing methods (as shown figure 3): (a) extraction of terms, (b) discovery of synonyms, (c) discovery of concepts, (d) generation of taxonomies, (e) generation of relationships (non-hierarchical) and (f) extraction of rules. According to the author, the steps can be viewed as the upper layers which requires more complex techniques of learning

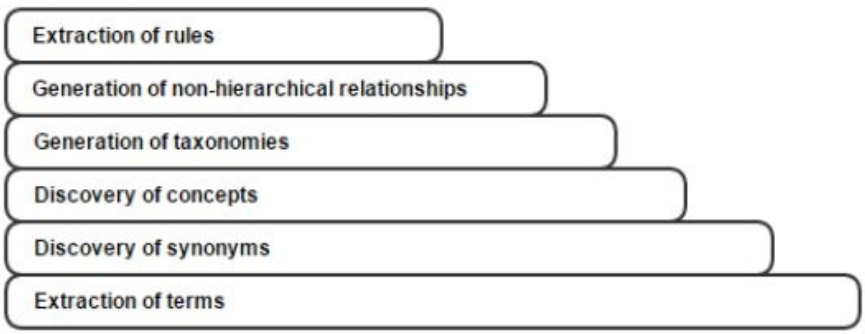

Fig. 3 Process for the creation of an ontology from text
Fonte: Buitelaar (2005)

4.0 PROPOSED METHODOLOGY

The web and digital social networks have changed the paradigm of products consumption. The previous generations of consumers did not had information resourses as large and easily accessible who the current generations has available, for example. However, However, according to a survey reported in Antohny (2016) 58% of Americans who use the Internet to search for product information before making their purchases, complain that information of this type is confusing and difficult to be found. From a business perspective, doing a manual analysis of criticism about a particular product recently released can take months, and therefore affect the decision-making process. Therefore, we propose will be to extract aspects of opinions using NLP methods and after to formalize this knowledge in an ontology. Segundo Gruber (1993) an ontology is responsible for formalizing shared concepts. For opinion mining, an ontology of product aspects shows the characteristics of the products making it accessible so that algorithms can identify users is opinion about aspects of the products evaluated on the web. For users and consumers of web products, having access to another user is review about a product you want to purchase is of great importance. That is the kind of information that influences on decision making, at the time of purchase. This research presented a methodology for the creation of an ontology of product aspects for e-business services. The ontology was implemented in OWL language. By using a set of opinionated texts containing user opinion about web products, we found that the formalization of knowledge about aspects of the evaluated products can enhance the e-business services. In addition, in the context of the semantic web, it is a necessary activity so that software agents can capture that information and make it available to users for decision-making at the time of purchase at the time of purchase. For ontology development are applied several methods. According to Bourigault (2002) one of the methods usually applied for extracting terms from the ontology, is using NLP techniques. In this work, four

techniques were used: *frequency , co-occurrence, morphosyntactic tagging and semantic clustering* and nine steps, according to figure 4.

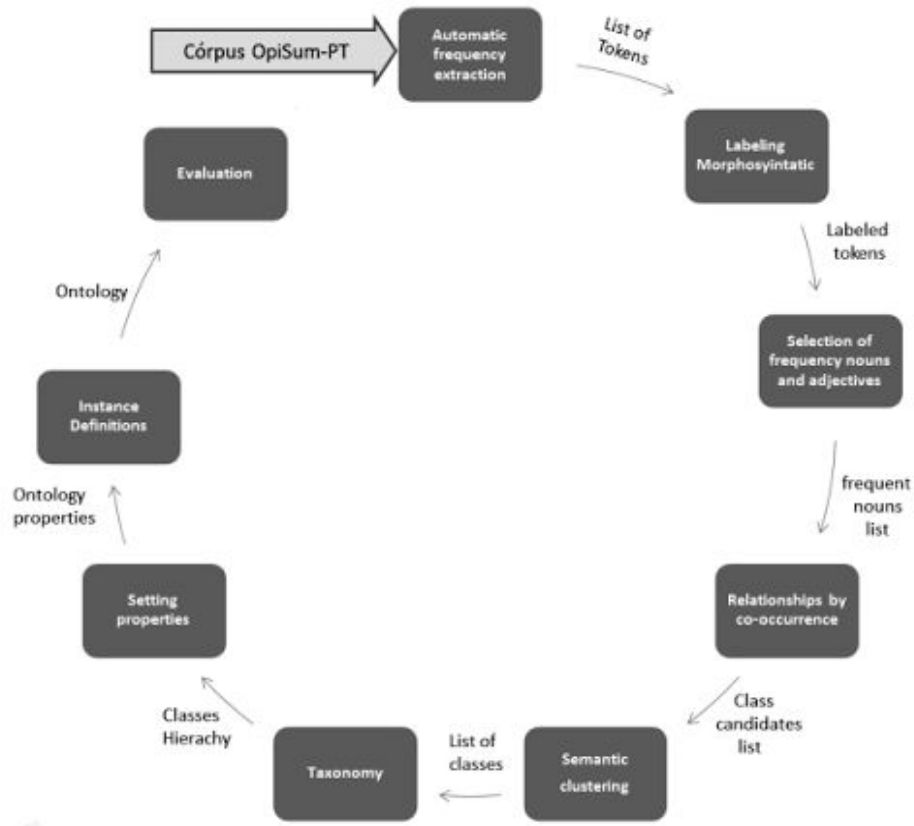

Fig. 4 Process for semi-automatic ontology development
Fonte: Vargas, F. A. (2016)

In this work will be using the frequency and co-occurrence of terms was performed through the Anticonc application (Anthony, 2016). For corpus morphosyntactic tagging use the TreeTager algorithm (Garcia and al, 2014) for the Portuguese. Then we used the Thesaurus Tep 2.0 (Mazieiro and al, 2008) for the clustering of terms as criteria for semantic relationship. Our goal with the application of this methodology is to simulate the creation of an ontology of semi-automatic way, from a corpus, in order to map the e-commerce products aspects evaluated by web users. We chose to use for the construction of ontology, the corpus domain OpiSums-PT Condori (2015). This corpus is a set of web users reviews on the web site Buscapé products. Once the processing of the corpus was finished, the ontology

construction process was conducted in OWL language by using the version of the software Protégé 4.3.

5.0 DEVELOPMENT AND RESULTS

The creation of an ontology is not an easy task; it requires a series of processes beyond the comprehension of the domain that you want to map. According to Gruber (1993) an ontology is a formal and explicit specification of shared concepts. Therefore, an ontology is an explicit knowledge of a domain through concepts, relationships, properties of concepts and constraints. The proposal of semi-automatic ontology creation is important because it minimizes the time spent identifying the most relevant terms and returns less subjective results, in order to maximize results. In this work, we propose the creation of an ontology owl through opinionated texts about web products, semiautomatic way. The development of this work can be divided into 9 (nine) processes, hierarchy as described in Figure 4. The texts used for processing are part of the córpus OpiSums-PT Condori (2015). Twenty documents were used to represent, respectively, two entities: *LG Smart-TV* and *TV-Smart-Samsung*. So they were processed in 1505 tokens and 605 types, as shown in Table 1.

Corpus	Entity	Number of Documents	Tokens	Types
OpiSums-PT	TV-Smart-LG; TV-Smart-Samsumg	20	1505	605

Table 1: Information corpus
Fonte: OpiSums-PT Condori (2015)

The processing of words or *tokens,* frequency morphosyntatic extraction and labeling was performed automatically using the first to the tokenization, Anticonc application (Anthony, 2016). It was then used for the labeling morphosyntactic Tree Tager algorithm (Garcia and al, 2014) for the portuguese. According Pang and Lee (2008), the most frequent nouns are usually the candidates for aspects. In total, 605 were labeled lexical terms. For the

selection of candidates for aspects, nouns groups were analyzed, adjectives, verbs and adverbs. In Figure 5 shows the amount of lexical gift items in the analyzed corpus, as his morphosyntactic label.

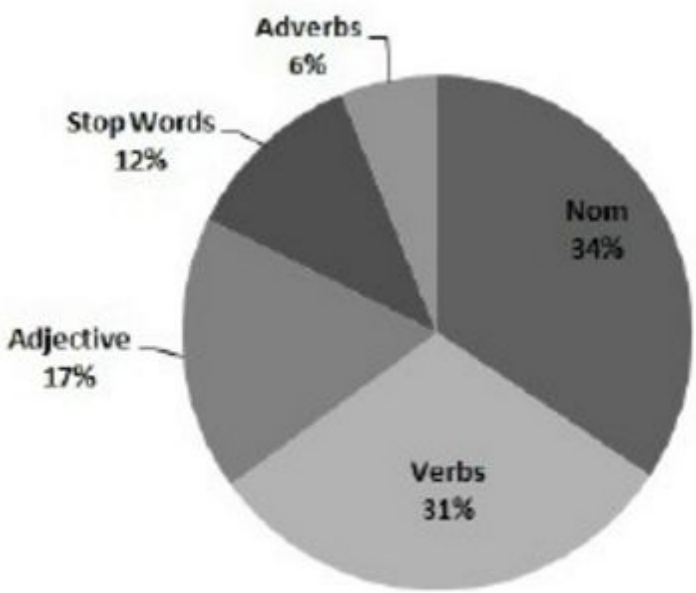

Fig.5 Information after labeling the corpus
Fonte: OpiSums-PT Condori (2015)

After morphosyntactic *tagger* have returned the grammatical class of terms, we separated only two groups found: nouns and adjectives. Nouns, they are candidates for aspects and adjectives they are candidates for feeling words. With the group of synonyms it was held clustering of terms that had synonymous relationship. A synonym relationship is manifested when there are two expressions that, for all contexts can substitute one for the other, without changing the meaning of the sentence (Lyons, 1977). For the clustering of terms that had synonymous relationship, we use the Thesaurus Tep 2.0 (Mazieiro and al, 2008)

From the topics generated by clustering, the co-occurrence analysis of these topics was held. The objective of this topic is to identify compound words. For example, the terms

“reprodução de áudio” (audio reproduction) and “reconhecimento de voz” (voice recognizing), it is intuitively possible to infer that they are compound terms or n-grams. The criteria used to determine whether a term was made up or not occurred from its neighbors. If neighbors are the noun type, it was considered a compound term.

To build the taxonomy, it was analyzed the relations between the entities. In the work the relations found were the type *part-of* . According to Winston and all (1987) the relationship expressed as *part-of* can be grouped into 6 (six) types of meronymy relationships: (I) component object, for example, the wing is part of the cup. (II) Member collection, for example, a tree is part of a forest. (III) portion-mass, for example, is that slice of the pie. (IV) raw object, for example, alcohol is Martini component. (V) task activity, for example, pay is part of buying. (VI) place-area, for example, the Everglades are part of Florida. Below the full taxonomy containing the properties and instances (Figure 5).

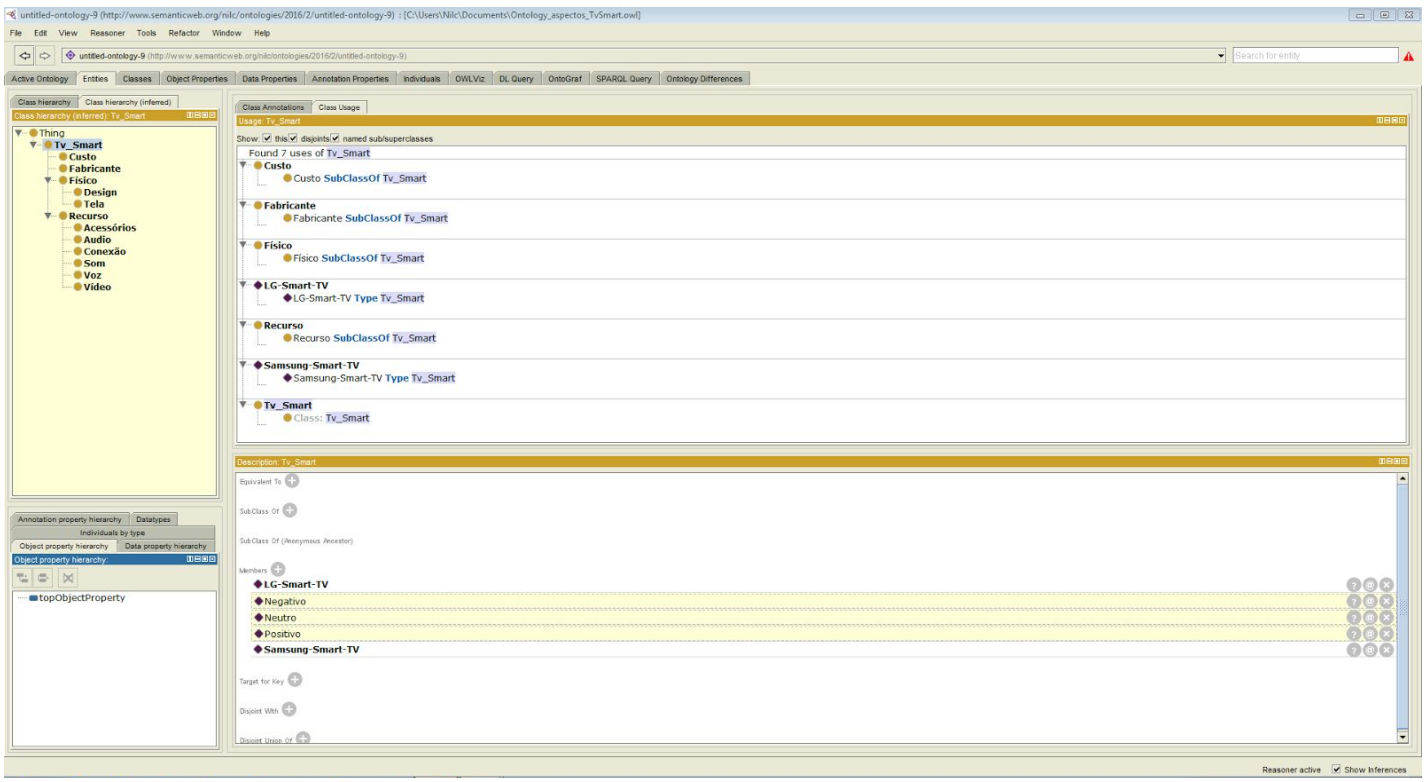

Fig.6 Taxonomy of products aspects
Fonte: Protégé 4.3

The final version has the ontology using the OWL language implemented, using the tool **Protégé 4.3.** And, the final graphical representation of the ontology features of web product is shown as Figure 6.

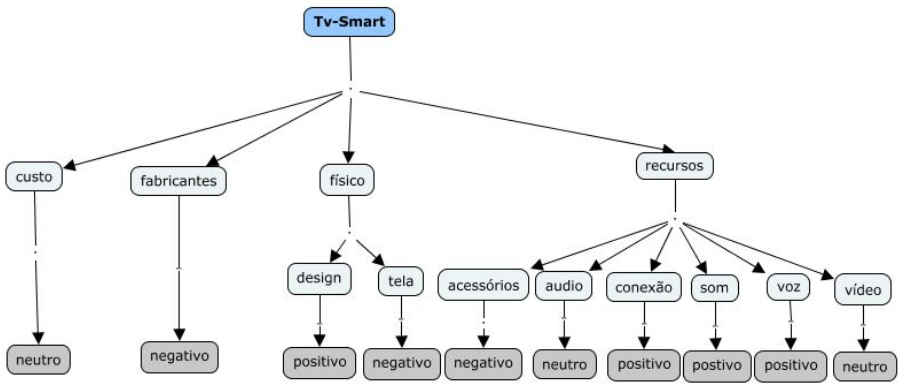

Fig.6 Ontology aspects of web products
Fonte: Vargas, F. A. (2016)

5.1 Discussion of the results

In the advance of web 2.0, the users in addition to produce content in the network and to interact with this content, its to produce semantic content. Recovery content semantic is big challenge. The construction of ontologies is a way to solve this problem. In the current web or web semantic proposed by Tim Berners-Lee, the content web is tracing in several modules divided is between mark-up languages, triple logic, inference engines and ontologies. The goals is to mapping semantic content, but it's not simple. For example, the construction of ontologies is not task easily because muny resources are necessary. In this work, we goal were to present a methodology for semi-automatic constrution of ontologies using PLN methods. We belive that the refined methods to constrution of ontologies way automated can get better the e-business processing. In the current web context many information is available on the web but litte knowlegde. To use ontologies can be used to formalision of the understanding of the Network.

In this work, we use NLP techniques for the construction of semi-automatic ontology. The application domain chosen was the business web services and e-business. From web users reviews, on electronic products, we have identified the target the features / aspects evaluated by users and formalized this knowledge in an ontology. According Guarino (2009) conceptualization is an abstract, simplified view of the world That We wish to represent for some purpose. Our approach differs to the previous approaches, because instead of just we represent the knowledge domain, use a fine level of granularity analysis of opinions. For this drink of concepts from mining of opinion and sentiment analysis. According to Liu (2012) opinion mining and sentiment analysis based on aspects is the level of analysis further analysis.

An ontology aspects formalizes the knowledge of the user opinion on the fine level of granularity. For example, in the section "I enjoyed my smartphone keyboard, but the battery is bad." In an analysis on a superficial level of analysis, the smartphone entity would be valued at but not their characteristics separately. The level of things, eh can capture the view on every aspect of the user regarding a product / entity. Therefore, product aspects of ontologies different from an ontology of opinions about products, provides a greater depth of analysis. This type of information is important for both electronics users, because they provide a set of summary information to aid in decision making, time to buy new. It is also important to manufacturers, which from an ontology of aspects of this kind can have a rapid return of the weaknesses of their products and solve the problem in a more agile way, decreasing sales. opinions aspect Ontologies also serve as one model summarized and structured opinions.

6.0 CONCLUSIONS AND FUTURE WORKS

6.1 Conclusions

The method of semi-automatic ontology creation using NLP methods as such criteria of *co-occurrence, frequency*, *semantic* and *morphosyntactic tagging* and *clustering* resulted in a cover of the equivalent domain 93%. Therefore, the semi-automatic creation of ontology using NPL criteria is satisfactory. The selection of the most common terms, returned only 2

(two) terms that were not part of the domain: *home* and *vice*. And, only 1 (one) term was not captured by the frequency method. The term *sound*, was part of the domain and had apparent frequency .

6.2 Future works

For future work, improving the coverage of the area we hope that the continuation of use of hybrid criteria, as well as using this work in tandem with the level of the sentence and not the lexical item can bring even better results. For the hybrid analysis, the level of the sentence considered linguistic patterns as a noun, adjective or noun-preposition-noun and applied statistics concomitantly may improve the results. The computational treatment in the process of ontology creation can bring great benefits while minimizing time and maximizing results. Our goal in this experiment was to present an ontology to represent the group aspects of web products from the assessment of web users consumers. We can conclude that the the opinion mining and semantic web applied to e-business services optimized the process of publishing information about the aspect of product reviews by the web users. First, because this kind of information is very relevant to consumer users in the purchase decision-making; secondly because the formalization of this content in OWL language allows search engines to capture this information as soon as possible by publicly improving the process of taking decision, when buying a new product. In addition, manufacturers of this product can measure how positive or negative aspect of a product is evaluated by consumers. The results can be demonstrated using both mining search field of view as the semantic web. For opinion mining, obtain a hierarchy of aspects is essential to identify the feeling attributed the opinions, which according to appearance-based approach, evaluations of web users are held on product feature. For the semantic web domain, formalize the web content to make it available so that machines can process it and search engines find it extremely valid. And finally, for e-business service the summarization of opinions about web products to improve the business process and can improve sales processes web product. Building an ontology is not an easy task, but according to some authors, it is the future of the internet. Some authors make this future projection, that is, for experts, for every university, company, business or government agency, there will be an ontology that describes.

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